

Q1

```
#define DEBUG 0
#if DEBUG
PRINT intermediate value
#endif
```

Q2

- **Fill constants/formula:**

```
#define PI 3.14159
DECLARE radius AS float
DECLARE area AS float
INPUT radius
area = PI * radius * radius
PRINT area
```

Q3

- **Output:** 15.000000 (or 15.0)
 - **Why float:** because result is declared as float and RATE is 1.5 (float)
-

Q4

- **Logical error:** condition should check MAX for upper bound; temp > MAX is wrong given your “valid range” intent
- **Correct condition:**

```
IF temp >= 0 AND temp <= MAX
PRINT "Valid"
ELSE
PRINT "Invalid"
```

Q5 (correct order)

```
DEFINE conversion rate
INPUT amount
```

```
CALCULATE converted value
PRINT result
```

Q6

```
IF (value >= MIN AND value <= MAX)
PRINT "Within range"
ELSE
PRINT "Out of range"
```

Q7

```
#define ENABLE_DISCOUNT 1
#define ENABLE_TAX 1

#define DISCOUNT_RATE 0.10
#define TAX_RATE 0.05

DECLARE price AS float
DECLARE quantity AS int
DECLARE subtotal AS float
DECLARE discount AS float
DECLARE tax AS float
DECLARE total AS float

INPUT price
INPUT quantity

subtotal = price * quantity
discount = 0.0
tax = 0.0

#if ENABLE_DISCOUNT
discount = subtotal * DISCOUNT_RATE
#endif

#if ENABLE_TAX
tax = (subtotal - discount) * TAX_RATE
#endif
```

```
total = (subtotal - discount) + tax
PRINT total
```

Q8

```
#define TAX 0.05
```

```
DECLARE price AS float
DECLARE total AS float
```

```
INPUT price
```

```
total = price + (price * TAX)
PRINT total
```

Q9

```
#define MODE 1 // 1: C to F, 2: F to C
```

```
#define PI2 2.0
```

```
#if MODE == 1
```

```
// Celsius to Fahrenheit:  $F = (C * 9/5) + 32$ 
```

```
#define A 9.0/5.0
```

```
#define B 32.0
```

```
#elif MODE == 2
```

```
// Fahrenheit to Celsius:  $C = (F - 32) * 5/9$ 
```

```
#define A 5.0/9.0
```

```
#define B -32.0
```

```
#endif
```

```
DECLARE inputTemp AS float
DECLARE outputTemp AS float
```

```
INPUT inputTemp
```

```
#if MODE == 1
```

```
outputTemp = (inputTemp * A) + B
```

```
#elif MODE == 2
outputTemp = ((inputTemp + B) * A)
#endif
```

```
PRINT outputTemp WITH 2 DECIMAL PLACES
```

Q10

```
#define SIZE 5
```

```
DECLARE arr [SIZE] AS float
DECLARE i AS int
DECLARE sum AS float
DECLARE avg AS float
```

```
INPUT arr elements
```

```
sum = 0.0
FOR i = 0 TO SIZE-1
sum = sum + arr[i]
END FOR
```

```
avg = sum / SIZE
```

```
PRINT sum
PRINT avg
```